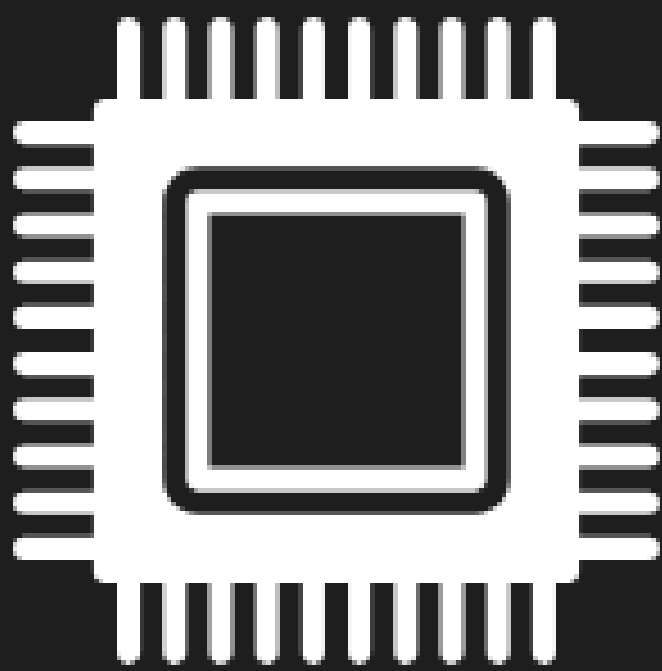




FLOWGRID

DESIGNED FOR NORMAL READY FOR DISASTER

BY

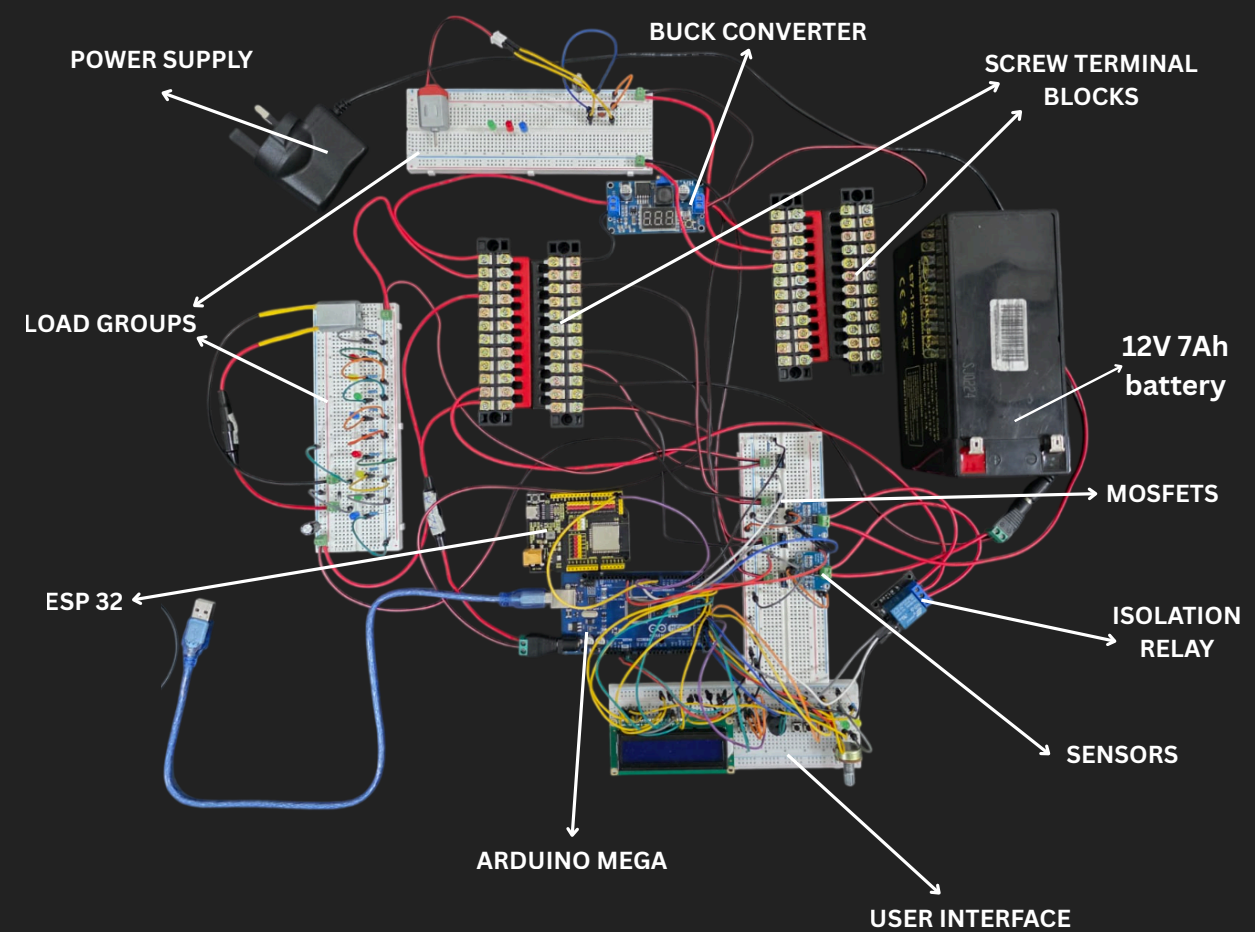


MICROLINK

WHAT IS FLOWGRID?

FlowGrid is an innovative project built by combining Raspberry Pi, ESP32, and Arduino Mega boards in a new way. By using the strengths of each platform, the MicroLink team developed a unique system for smart power management. Flow grid is the result of the passion and hours of dedication by all the members at micro link. It uses intelligent micro power grid that provides reliable, prioritized electricity during blackouts, grid failures, and natural disasters. It constantly monitors grid conditions and danger levels, automatically switching between available AC and DC power sources including the main supply, renewable energy, and stored power to keep essential systems running. An integrated system distributes power based on priority needs. Hospitals, emergency services, and communication networks receive maximum power, while non-essential uses such as home air conditioning and entertainment systems are reduced or turned off. The system uses a danger score calculated from real-time sensor data to determine its operating state. The AI does not directly control loads; it acts as an advisory layer that adjusts the danger score by incrementing or decrementing it based on past events and patterns. Over time, this improves response accuracy and reliability. Designed for both smart cities and developing countries.

LABELLED DIAGRAM



Component	Cost / AED
Mega	75
Screw	40
Buck	15
Wires	40
Relay	5
Battery	50
Breadboard	20
Sensors	50
MOSFETs	50
Adapter	25
Fuses	25
ESP	50
Total	445

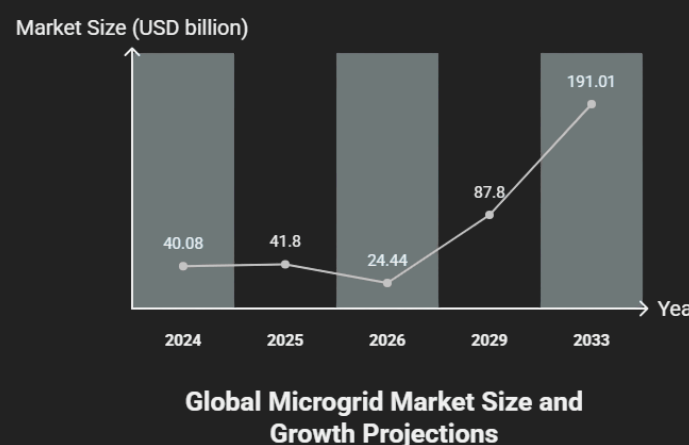
SCAN FOR CODE



- **MARWAN KHALED**
 - +971 55 391 9637
- **SHRESHTH DADHICH**
 - +971 58 185 4065
- **AFAN NOMANI**
 - +971 50 817 6014
- **MARAGATHA SUNDARAM**
 - +971 56 785 9265
- **MUHAMMED FAIQ**
 - +971 50 989 8028

OUR COSTS

- The first prototype of FLOW was built using several different components bought from a mix of local markets and online stores, all at normal retail prices. Producing this prototype in-house cost us around 500 AED in total. This higher cost mainly reflects the fact that it was a one-off build, with no bulk discounts and a lot of manual assembly involved. As we move toward larger-scale production, this cost will drop significantly. By purchasing components in bulk directly from suppliers and factories, and by setting up a proper assembly line, we can greatly reduce both material and labor costs.
- Overall, we expect these changes to cut production costs by at least 50 percent. With these improvements in place, Micro Link will be able to produce and sell FLOW units for as little as 100 AED each. This will allow us to reach not only urban customers, but also people who are looking for well-made products at an affordable price.



HARDWARE DETAILS

- At the core of the system is an Arduino Mega, chosen for its high input/output capacity and expandability. The controller continuously measures voltage and current across multiple load branches using dedicated current and voltage sensing modules, providing real-time, branch-level insight into power consumption rather than relying on aggregate system measurements.
- Each load branch is controlled using low-side N-MOSFET switching, enabling the system to disconnect, protect, or prioritize loads dynamically based on operating conditions. Critical loads can be exempted from throttling logic to ensure uninterrupted operation during faults or energy shortages. Hardware-level safety is reinforced through fast-blow fuses, ensuring failures remain localized and do not cascade through the microgrid.
- FlowGrid operates in three modes—Normal, Conserving, and Disaster—which automatically adjust load priorities and shed non-essential loads during periods of limited energy availability. A 16x2 LCD interface with physical navigation buttons provides clear, real-time system feedback without requiring external software or connectivity.
- While demonstrated as a DC prototype, FlowGrid is intentionally designed to integrate DC-AC inverters, AC distribution stages, and bidirectional converters, enabling seamless expansion into hybrid microgrids that combine renewable generation, storage, and conventional AC loads at larger scales.

SOFTWARE DETAILS

- The team in charge of coding life into our carefully designed product has used multiple languages such as C++, C and Python to provide the most efficient, smart and dynamically adaptive code that helps the FLOW not only run faster and be the best at what it was built for. It also helps save resources used to build the FLOW as the code only needs a storage of 200MB to run flawlessly throughout its lifespan.
- The code is made such that any user can easily set different priority levels to each pin on the boards and choose which systems they consider to be more important during an outage.
- The Flow will also receive regular software and security updates when connected to the internet in order to keep it safe and functioning. The code also allows for communication between the different types of boards that make up the flow.

LEGAL AND REGULATORY REQUIREMENTS

- MICROLINK has made sure to factor in any UAE regulations involved with the connection of FLOW to the actual commercial grid supplying the country.
 - Federal Decree by Law No. (17) of 2022 Regulating the Connection of Renewable Energy Production Units Distributed on the Electrical Grid
 - Cabinet Resolution No. (103) of 2022 Concerning the Administrative Penalties for Acts that Violate the Provisions of Federal Decree-Law No. (17) of 2022 Regulating the Connection of Distributed Renewable Energy Production Units to the Electrical Grid

